

NYC Taxi Data Analysis & Insights

PROJECT OBJECTIVE • INDEPENDENT ANALYTICS PROJECT

Project Period

03/08/2025 to Present

Executive Objective

Develop a data-driven view of New York City yellow-taxi operations by analyzing public trip records to understand demand, pricing, trip behavior, payment activity, and tipping patterns. The analysis is designed to turn operational data into clear business insights that can support planning, service improvement, and performance monitoring.

Primary Analytical Objectives

- 01

Demand & Timing

Identify peak periods by hour, day of week, month, and other time-based patterns.
- 02

Trip Economics

Measure relationships between trip distance, fare amounts, duration, surcharges, and total customer cost.
- 03

Customer Behavior

Evaluate passenger counts, payment methods, and tipping behavior across taxi trips.
- 04

Location Intelligence

Identify high-activity pickup and drop-off zones and examine movement patterns across New York City.
- 05

Data Quality

Detect missing values, invalid records, extreme distances, unusual fares, and other anomalies before analysis.
- 06

Decision Support

Convert analytical findings into practical recommendations for operations, planning, and service management.

Expected Business Outcome

The final project will produce a professional analytics portfolio showing how taxi-trip data can be cleaned, explored, visualized, and translated into actionable insights. Outputs will include an analytical notebook, visualizations, documented methodology, and a GitHub-ready project repository.

Data Source & Project Status

This project uses publicly available NYC Taxi & Limousine Commission (TLC) trip-record data. TLC states that yellow-taxi records contain pickup/drop-off times and locations, trip distance, itemized fares, rate types, payment types, and driver-reported passenger counts. The dataset is published by TLC and is not presented here as proprietary company data.

Project classification: Independent data-analysis portfolio project. Not an official publication of the NYC Taxi & Limousine Commission and not affiliated with or endorsed by the City of New York.

Source: NYC Taxi & Limousine Commission — TLC Trip Record Data.